Digital Transformation KPI Dashboard

KPI Rationale and Strategic Interpretation, FY2021–FY2025

Basis of analysis. Deere & Company is publicly traded (NYSE: DE). This analysis draws on its Form 10-K and Form 8-K filings for fiscal years 2021–2025, its investor materials and its own sustainability disclosures. R&D intensity and FY2024 operating margin are calculated from disclosed figures. All other dashboard values are reported figures or rounded milestone disclosures. Calculated values are marked *. Adoption KPIs come from Deere's 2024 Sustainability Disclosures (FY2021–FY2024) and its 2025 Business Impact Report (FY2025), so every year is disclosed and nothing is interpolated. No private or fabricated data is used.

Team Members & Roles

Member	Role	Contribution
Chandrahas Reddy Bheemireddy	Metric Selector	Defined Deere's transformation thesis and selected the eight KPIs with a leading and lagging balance. Wrote the KPI rationale and the strategic interpretation.
Sai Nikhitha Kasala	Data Analyst	Sourced and verified the public-filing data behind every KPI. Documented the calculations, assumptions and inflection points.
Swapna Tumma	Dashboard Builder	Built and tested the self-contained HTML + Chart.js dashboard. Added the inflection-point annotations and the baseline-versus-latest summary.
Venkatanagasai Tanish Puttagunta	Co-Data Analyst / QA Support	Cross-checked sources and reviewed consistency between the dashboard and this report. Supported the strategic interpretation and prepared the submission package.

Live dashboard: https://nikhitharediee12.github.io/hilti-transformation-roadmap/week7/week7_group5_Deere%20%26%20company_KPI_Dashboard.html

KPI Rationale

Deere's transformation thesis is that the company is moving from a cyclical equipment business toward a technology-enabled "Smart Industrial" model in which connected machines, precision technology and software raise customer value and make earnings more resilient. Deere's filings describe a technology stack and lifecycle solutions built around customers' production systems, and Leap Ambitions, introduced in 2022, set adoption targets for 2026. The KPI set therefore has to separate evidence that Deere is building the capability from evidence that the capability is changing financial outcomes. Deere does not publish a consistent annual digital-revenue series, so the dashboard combines disclosed financial metrics with the three adoption measures Deere itself reports (connected machines, engaged acres and highly engaged acres), each of which is disclosed for every year from FY2021 to FY2025.

First, R&D intensity is the leading indicator of commitment. Intensity (R&D divided by equipment operations net sales) rose from about 4.0% in FY2021 to about 5.9% in FY2025. It matters most in the downturn: R&D spend kept rising while sales fell, so intensity jumped, which shows Deere protected its technology budget when it could have cut it. Absolute spend ($1,587 million in FY2021 to $2,311 million in FY2025) uses the same numerator as intensity, so it is kept in the annotations and interpretation rather than as a separate KPI.

Second, connected machines, engaged acres and highly engaged acres are the most direct leading indicators of digital adoption, and each measures something different. Connected machines count the fleet feeding the John Deere Operations Center. Engaged acres count unique acres with at least one documented operation in the Operations Center

over 12 months, which shows the breadth of use. Highly engaged acres count engaged acres with documentation of multiple production steps and the use of digital tools for multiple value-creating activities, which shows the depth of use. Deere set 2026 targets of 500 million engaged acres (50% of them highly engaged) and 1.5 million connected machines, so all three can be tracked against plan. FY2025 values are Deere's rounded figures (nearly 500 million engaged acres and 1 million connected machines), so FY2025 changes are approximate.

Third, net sales, operating margin and net income are lagging value indicators. Together they show whether the transformation is converting into earnings, and they expose the farm-equipment cycle, which is the main confounder. Net sales rose from $39.7 billion in FY2021 to a $55.6 billion peak in FY2023 and then fell to $38.9 billion in FY2025, while operating margin moved from about 17.3% to a 21.9% peak and then to 12.6%. The FY2024 margin is computed from the three segments' disclosed operating profit and is labeled as calculated wherever it appears.

Fourth, the cost of sales ratio is a lagging operating-leverage indicator. It fell from 73.3% in FY2021 to 67.9% in FY2023 and rose to 72.4% in FY2025, which helps separate production-cost and pricing effects from the investment and tariff effects that sit below gross margin. Stock price is excluded because it reflects market expectations rather than outcomes management controls. The set has four leading and four lagging indicators, so investment activity is never treated as success by itself. On the dashboard, net sales is a stacked bar by segment so mix shifts are visible, net income is a bar because it is a period comparison, and the ratios and adoption measures are lines because they are trends.

Strategic Interpretation

The data suggests that Deere is building digital capability, but the financial payoff is not yet visible in margins: adoption and investment kept rising through a downturn while profitability fell back below its FY2021 level. The first inflection came in FY2022, when Deere introduced Leap Ambitions and R&D rose 20% to $1.9 billion; net sales grew 21% while operating margin stayed near 17.4%, so the extra investment was absorbed by cycle growth without lifting margin. The second came in FY2023, the cycle peak, with net sales of $55.6 billion, operating margin of 21.9% and net income of $10.2 billion. Deere attributes the gains to price realization and strong volumes, so the peak says more about the market than about technology.

In FY2024, net sales fell 19% but R&D still rose to $2.3 billion, lifting R&D intensity from 3.9% to 5.1%, so Deere protected technology spending. In FY2025, net sales of $38.9 billion nearly match FY2021's $39.7 billion, yet operating margin is 12.6% against about 17.3% and net income is 16% lower. R&D is $724 million higher than in FY2021 and management cites about $600 million of tariffs, roughly 1.5 points of sales. Adoption reached nearly 500 million engaged acres and about 1 million connected machines (short of the 1.5 million goal), with 147 million acres (roughly 30%) highly engaged against the original 50% goal. Deere's adoption indicators are advancing faster than the publicly visible evidence of monetization. Current disclosures do not establish whether increased platform usage is producing recurring revenue or protecting margins, the key question for management as R&D intensity stays high.

KPI Data Table

KPI	FY2021	FY2022	FY2023	FY2024	FY2025
Net sales, equipment operations ($m)	39,737	47,917	55,565	44,759	38,917
Operating margin (%)	17.3	17.4	21.9	18.2*	12.6
Cost of sales ratio (%)	73.3	73.7	67.9	68.8	72.4
Net income attributable to Deere ($m)	5,963	7,131	10,166	7,100	5,027
Net sales: Production & Precision Ag ($m)	16,509	22,002	26,790	20,834	17,311
Net sales: Small Ag & Turf ($m)	11,860	13,381	13,980	10,969	10,224
Net sales: Construction & Forestry ($m)	11,368	12,534	14,795	12,956	11,382
R&D intensity (%)*	4.0	4.0	3.9	5.1	5.9
Memo: R&D spend ($m), not a KPI	1,587	1,912	2,177	2,290	2,311
Connected machines (K)	444	500	650	775	1,000
Engaged acres (M)	315	329	388	455	~500
Highly engaged acres (M)	66	68	92	125	147

Notes. * calculated. R&D intensity and FY2024 operating margin are calculated from disclosed figures. All other dashboard values are reported figures or rounded milestone disclosures. The three segment rows add up to equipment operations net sales and are the stacks in the dashboard chart. FY2024 operating margin is the sum of the three segments' disclosed operating profit ($8,150M) divided by net sales; FY2021 ($6,868M / $39,737M), FY2022, FY2023 and FY2025 are disclosed. R&D intensity = R&D expense / equipment operations net sales. Cost of sales ratio = cost of sales / net sales (as stated in Deere's filings). Adoption values for FY2021–FY2024 are from Deere's 2024 Sustainability Disclosures and for FY2025 from its 2025 Business Impact Report; no values are interpolated. FY2025 engaged acres is Deere's "nearly 500M" (shown as ~500) and FY2025 connected machines is Deere's rounded 1M, so FY2025 change figures for these two KPIs are approximate. Highly engaged acres are engaged acres with documentation of multiple production steps and the use of digital tools for multiple value-creating activities over 12 months. R&D spend is a memo row (same numerator as R&D intensity) and is not a KPI. Deere's fiscal year ends in late October or early November.

Key Inflection Points

- **FY2022:** R&D rose 20% to $1.9B as Leap Ambitions was introduced, with 2026 targets of 500M engaged acres and 1.5M connected machines; margin stayed near 17.4% on 21% sales growth.
- **FY2023:** net sales peaked at $55.6B, operating margin at 21.9% and net income at $10.2B, driven by price realization and strong volumes; engaged acres rose to 388M and highly engaged acres to 92M.
- **FY2024:** net sales fell 19% while R&D rose to $2.3B, lifting R&D intensity from 3.9% to 5.1%.
- **FY2025:** margin fell to 12.6% and net income to $5.0B (about $600M of tariffs), while adoption reached nearly 500M engaged acres (147M highly engaged) and about 1M connected machines against a 1.5M-machine goal.

Sources & Links

Source	Data used	Link
Deere FY2021 Form 10-K	FY2021 segment net sales and operating margins (basis for the FY2021 operating margin of 17.3%).	https://www.sec.gov/Archives/edgar/data/315189/000155837021016864/de-20211031x10k.htm
Deere Q4 FY2022 Form 8-K (Nov 23, 2022)	FY2022 and FY2021 net sales, net income and segment margins.	https://www.sec.gov/Archives/edgar/data/315189/000155837022018162/de-20221123xex99d1.htm

Sources & Links — continued

Source	Data used	Link
Deere FY2023 Form 10-K	FY2021–FY2023 net sales, cost of sales, R&D and net income; Leap Ambitions.	https://www.sec.gov/Archives/edgar/data/315189/000155837023019812/de-20231029x10k.htm
Deere Q4 FY2023 Form 8-K supplemental data (Nov 22, 2023)	FY2022 and FY2023 equipment operations operating profit as % of net sales.	https://www.sec.gov/Archives/edgar/data/315189/000155837023019292/de-20231122xex99d2.htm
Deere FY2023 Annual Report to Shareholders	Cost of sales ratio 67.9% vs 73.7%; price realization and volume drivers.	https://www.sec.gov/Archives/edgar/data/315189/000155837024000174/de-20231029xars.pdf
Deere Q4 FY2024 earnings call slides (Nov 21, 2024)	FY2024 segment net sales and operating margins (cross-check of the FY2024 margin, which is computed from the FY2025 Form 8-K).	https://www.sec.gov/Archives/edgar/data/315189/000155837024015876/de-20241121xex99d2.htm
Deere FY2024 Form 10-K	FY2022–FY2024 net sales, cost of sales, R&D and net income.	https://www.sec.gov/Archives/edgar/data/315189/000155837024016169/de-20241027x10k.htm
Deere FY2025 Form 10-K	FY2025 results, 72.4% cost of sales ratio, about $600M tariff impact, Leap Ambitions, and the risk factor on slower-than-expected adoption of precision and subscription offerings.	https://www.sec.gov/Archives/edgar/data/315189/000110465925122321/de-20251102x10k.htm
Deere Q4 FY2025 Form 8-K (Nov 26, 2025)	FY2025 and FY2024 net sales, segment net sales and operating profit, R&D expense ($2,311M, $2,290M) and net income ($5.027B).	https://www.sec.gov/Archives/edgar/data/315189/000110465925116130/de-20251126xex99d1.htm
Deere 2024 Sustainability Disclosures and Metrics	FY2021–FY2024 adoption series: engaged acres 315M / 329M / 388M / 455M; highly engaged acres 66M / 68M / 92M / 125M; connected machines 444K / 500K / 650K / 775K.	https://www.deere.com/assets/pdfs/common/our-company/sustainability/data-book-2024.pdf
Deere 2025 Business Impact Report (Jan 15, 2026)	FY2025 adoption: nearly 500M engaged acres, 147M highly engaged acres and 1M connected machines.	https://s22.q4cdn.com/253594569/files/doc_downloads/2026/Business_Impact_Report_2025.pdf
Deere 2022 Sustainability Report	Engaged acres 315M (2021) and 329M (2022) and highly engaged acres 66M and 68M, shown against the 500M-acre target on the same basis as later years.	https://www.deere.com/assets/pdfs/common/our-company/sustainability/sustainability-report-2022.pdf
Deere FY2024 Business Impact Report	455M engaged acres and 775,000 connected machines in FY2024 (Deere's own report, Nov 2023 to Oct 2024).	https://deere.com/sustainabilityreport
Deere 2021 Annual Report (Dec 16, 2021)	315M engaged acres in 2021; equipment operations net sales $39,737M and operating profit $6,868M (17.3%); cost of sales ratio 73.3%; R&D $1,587M.	https://s22.q4cdn.com/253594569/files/doc_financials/2021/ar/Deere-Co_Annual-Report-2021.pdf
Deere Investor Presentation (Feb 2022)	2026 targets: 500M engaged acres (50% highly engaged) and 1.5M connected machines.	https://s22.q4cdn.com/253594569/files/doc_presentations/2022/02/Investor-Presentation_2022_Feb-update.pdf
Digital Commerce 360, Deere investor day (Dec 11, 2025)	About 500M engaged acres and 1M+ connected machines. Cross-check of the FY2025 values; the primary source is Deere's 2025 Business Impact Report.	https://www.digitalcommerce360.com/2025/12/11/john-deere-emphasizes-digital-scale-and-connected-customers/